

Flavours of Internet Telephony

PC-to-PC



PC-to-Phone



Phone-to-Phone



1. PC-to-PC calls have been made for years; no telephone, just the next best thing
2. PC-to-Phone calls are routed through special links to the telephone network
3. Though not available in India yet, Phone-to-Phone eliminates the need for an expensive PC

INFOGRAPHIC: Jaya Shetty

of the call, both ends of the conversation are engaged, leaving no room for communication outside the link. Telephonic dialogues are transmitted across the network at a fixed speed of approximately 64 kbps in each direction, giving a total broadcast speed of 128 kbps. In a typical conversation, most of the bandwidth is underutilised as there are pauses and other moments of silence. In one-way conversations, at least half of the available bandwidth is wasted while the other party is listening.

How to use it

If you want to set up Net telephony, the first thing you need is a computer. To make sure that it's outfitted with everything you need, here's a list of the essential components.

- Pentium II class CPU running at 233 MHz
- 128 MB RAM
- 16-bit full-duplex soundcard
- Microphone headset combo
- The latest version of your favourite browser

If you don't have a PC, you can still make VoIP calls using an Internet Protocol (IP) device, which will cost you something in the rage of Rs 3,000 to Rs 15,000 depending upon the handset and features.

In case you lack both, just walk into a cyber cafe and make the calls you want. For instance, Satyam Infoway plans to provide services to those who may not have PCs at home. The ISP, which runs the I-Way chain of cyber cafes, plans to outfit 700 centres with VoIP services.

Say for instance, you are loading a rather lengthy Web page. No matter how quickly the site may load, not all of the site data is sent across at once. Rather, the sending computer (server) dispatches the data in small packets with the address affixed to each one telling the network where to send them. The receiving computer receives these packets and reassembles them to form the site in its entirety. The whole process is extremely efficient; multiple connections can be simultaneously made to several computers at once with packet switching. It also minimises the time for which a connection is maintained between computers and this greatly reduces the load on the network.

VoIP employs this packet switching method to deliver many pluses over circuit switching technology. A single dial-up Net connection allows for many more calls in the amount of space occupied by a single call in a circuit switched network. For instance, a

A typical five-minute conversation would translate into roughly 4,800 KB, or 16 KB for each second that the circuits remain open. Keep in mind that these figures apply irrespective of whether or not speech is being transmitted over the lines. Imagine being able to view only one Web page at a time—that's how today's telephone networks operate.

Data based networks such as those hosted on the Internet do not use this type of dedicated circuit switching. Instead, data networks use a completely flexible means of sending information called packet switching. While circuit switching keeps a connection open and constant, packet switching opens a connection barely long enough to send a chunk of data from one computer to another.

Here is what happens between two computers over the Internet.

PHOTOGRAPH: Mexy Xavier